Print guide

DIY CNC epoxy-granite mill · October 2026

Every printed part for the molds, the castings and the machine, cut to fit a Bambu A1 mini (180 mm cube; no piece is over 170 mm). The STL files are in `print/`, one folder per casting, with the list in `print/manifest.csv`. They are written by `model/build_cnc.py` on every run. How the molds go together is in the mold guide.

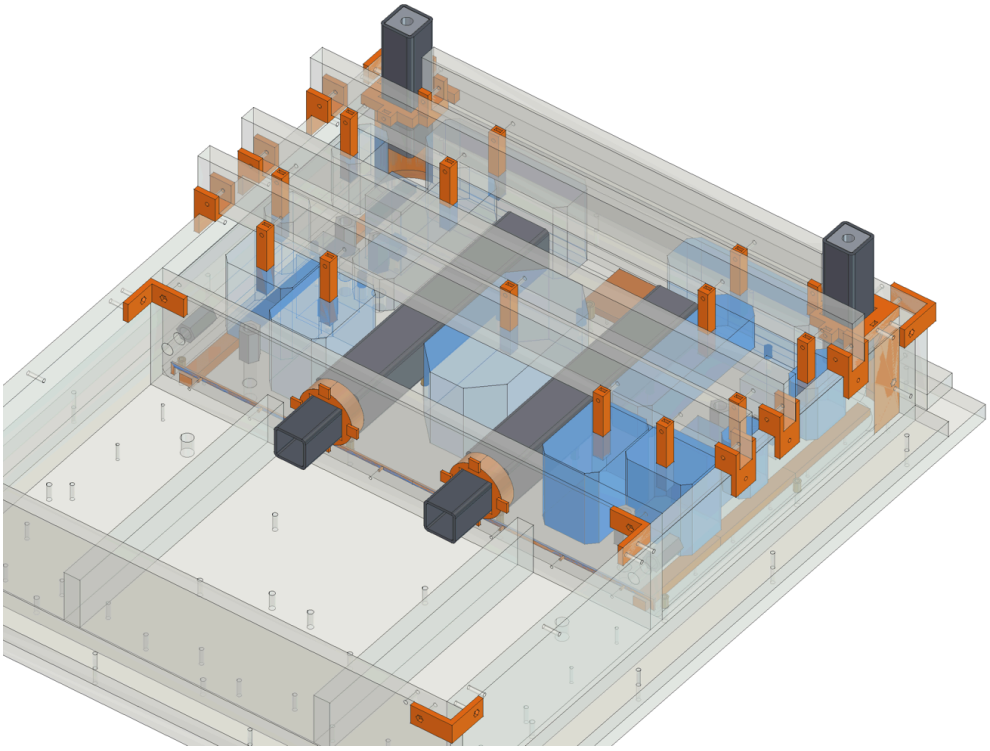
148 prints, 1.31 kg of PETG (two 1 kg rolls). The filament figures are sliced in Bambu Studio (A1 mini, 0.4 mm nozzle, 0.2 mm layers, PETG) with the settings below, by `model/slice_prints.py`.

Settings

- **PETG for everything.** Thick sections of curing epoxy get warm, and PLA softens at about 55 °C.
- 0.4 mm nozzle, 0.2 mm layers, 4 top and 3 bottom layers. Every file is already turned the way it prints best, without supports.
- **Cores** (`core-*`): thin shells, 1.2 mm walls (three lines), so they print solid; the posts get 2 walls and 10 % infill. The sand inside holds the walls while casting.
- **Fillers, formers, plugs** (`filler-*`, `former-*`, `plug-*`): 2 walls, 10 % infill. They only have to hold their shape against the mix; the nut pockets have their own walls.
- **Mold kit** (`kit-*`: brackets, saddles, holders, bushings) and machine parts: 3 walls, 15 % infill. Print one, check it over a 19 mm offcut of your board, then the rest.
- **Liners and cast-in parts** (`liner-*`, `cast-in-*`): 3 walls, 40 % infill.
- **Fits.** Slip fits are 0.15 mm a side, nut pockets 0.15 mm over the nut, pins 3.0 mm into a 1/8 in hole. If your printer runs tight, print a bracket and a former first and adjust the horizontal expansion (XY compensation) rather than the files.

Joining the pieces

- **Nuts.** Every piece that bolts to a board has its nut built in. On cast faces the nut drops into a pocket in the face against the board and slides sideways under the bolt hole; on brackets it presses into a hex pocket. Fit the nuts before the piece goes on the board.
- **Parts cut to fit the bed** (`-partNofM`) have a peg on one cut face and a socket on the other: push them together, no glue. Each part has its own bolts or pins to the board.
- **Cores** cut in two have a bulkhead on each side of the cut with two M3 x 10 bolts through them.
- **Pins.** Thin formers (gasket-groove beads, the shallow end of a gutter) have 3 mm pins that push into 1/8 in holes. Nothing else holds them; the casting presses them onto the board.
- **Removable pieces** (formers, fillers, plugs, kit): 3-5 coats of paste wax and PVA like the boards. **Pieces that stay in** (core posts, cast-in parts): no wax; scuff and degrease them so the epoxy grips.



Foam pieces

The base pocket cores and the socket cores are XPS foam board (`foam=True`, the default in `model/build_cnc.py`); only the pocket cores' posts and the socket spikes are printed. They are hidden and only need to be within a millimeter or two. The coolant trough, gutters and the beam's back edge print, because they shape faces you see and drain coolant (`foam_formers=True` cuts those from foam too). The mold guide lists the foam pieces with their sizes. `foam=False` prints the cores (about 1.4 kg more).

Tools: pour-line gauge and screed shoes

Folder `print/tools/`, about 29 g.

File	What	Qty	Size (mm)	g (all)	Notes
`kit-pour-gauge.stl`	Mold kit (reused)	2	25 x 29 x 24	9	Clips over a wall top; its inner leg ends 25 mm down, on the pour line. Pencil along it all round the mold, and check the fill against it.
`kit-screed-shoe.stl`	Mold kit (reused)	2	49 x 27 x 32	20	A pair: rest them on two facing wall tops and drop a straight board (on edge) into them; its bottom edge sits on the pour line. Slide it along to strike off.

Base mold kit (both halves)

Folder `print/base-box/`, about 187 g.

File	What	Qty	Size (mm)	g (all)	Notes
`kit-corner-bracket.stl`	Mold kit (reused)	4	50 x 50 x 23	33	
`kit-partition-bracket.stl`	Mold kit (reused)	2	44 x 44 x 23	14	
`kit-saddle.stl`	Mold kit (reused)	6	53 x 31 x 25	58	The rear half needs six; four move over for the front half.
`kit-leg-holder.stl`	Mold kit (reused)	4	72 x 72 x 40	82	Set the leg 92 mm proud of the holder and tighten the pinch screw (M4 x 20).

Base, rear half (cast first)

Folder `print/base-rear/`, about 245 g.

File	What	Qty	Size (mm)	g (all)	Notes
`former-gutter-L-part1of3.stl`	Former (removable)	1	15 x 124 x 12	8	Bolts to the floor from underneath (the base is cast top face down).
`former-gutter-L-part2of3.stl`	Former (removable)	1	15 x 124 x 10	8	
`former-gutter-L-part3of3.stl`	Former (removable)	1	15 x 120 x 9	7	
`former-gutter-R-part1of3.stl`	Former (removable)	1	15 x 124 x 12	8	Bolts to the floor from underneath (the base is cast top face down).
`former-gutter-R-part2of3.stl`	Former (removable)	1	15 x 124 x 10	8	
`former-gutter-R-part3of3.stl`	Former (removable)	1	15 x 120 x 9	7	
`former-motor-relief.stl`	Former (removable)	1	72 x 60 x 14	12	Bolts to the floor from underneath (the base is cast top face down).
`filler-corner-L.stl`	Filler (removable)	1	40 x 40 x 110	17	Rounded corner; bolts to the floor and both walls.
`filler-corner-R.stl`	Filler (removable)	1	40 x 40 x 110	16	Rounded corner; bolts to the floor and both walls.
`former-split-gasket-groove-part1of4.stl`	Former (removable)	1	145 x 14 x 8	2	Pins into the partition.
`former-split-gasket-groove-part2of4.stl`	Former (removable)	1	145 x 4 x 8	1	
`former-split-gasket-groove-part3of4.stl`	Former (removable)	1	145 x 4 x 8	1	

File	What	Qty	Size (mm)	g (all)	Notes
`former-split-gasket-groove-part4of4.stl`	Former (removable)	1	145 x 14 x 8	2	
`core-center-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-foot-b-L-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-foot-c-L-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-knee-L-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-outer-a-L-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-outer-b-L-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-rear-L-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-foot-b-R-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-foot-c-R-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-knee-R-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-outer-a-R-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-outer-b-R-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-rear-R-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`plug-socket-spike.stl`	Plug (removable)	4	14 x 14 x 20	8	Bolts to the floor; the foam socket block pushes over it (cut a 14 mm square hole 20 deep in the block).

File	What	Qty	Size (mm)	g (all)	Notes
`kit-spigot-bushing.stl`	Mold kit (reused)	2	80 x 80 x 22	34	Push fit in a 2-1/2 in hole-saw hole; flange on the far side.
`cast-in-base-rear-tube-plug-l.stl`	Cast-in part (stays in)	2	43 x 10 x 43	15	Cast in, stays.
`cast-in-base-rear-spigot-sleeve-l.stl`	Cast-in part (stays in)	2	43 x 69 x 43	22	Cast in, stays.
`cast-in-leg-rear-l-plug.stl`	Cast-in part (stays in)	2	31 x 31 x 10	8	Cast in, stays.

Base, front half

Folder `print/base-front/`, about 284 g.

File	What	Qty	Size (mm)	g (all)	Notes
`former-gutter-L-part1of2.stl`	Former (removable)	1	15 x 149 x 15	11	Bolts to the floor from underneath (the base is cast top face down).
`former-gutter-L-part2of2.stl`	Former (removable)	1	15 x 145 x 13	10	
`former-gutter-R-part1of2.stl`	Former (removable)	1	15 x 149 x 15	11	Bolts to the floor from underneath (the base is cast top face down).
`former-gutter-R-part2of2.stl`	Former (removable)	1	15 x 145 x 13	10	
`former-trough-part1of5.stl`	Former (removable)	1	116 x 35 x 19	20	Bolts to the floor from underneath (the base is cast top face down).
`former-trough-part2of5.stl`	Former (removable)	1	116 x 35 x 23	22	
`former-trough-part3of5.stl`	Former (removable)	1	116 x 35 x 33	25	
`former-trough-part4of5.stl`	Former (removable)	1	116 x 35 x 24	23	
`former-trough-part5of5.stl`	Former (removable)	1	112 x 35 x 21	21	
`filler-corner-L.stl`	Filler (removable)	1	40 x 40 x 110	21	Rounded corner, and the plug that forms the draw-rod nut pocket; bolts to the floor and both walls.
`filler-corner-R.stl`	Filler (removable)	1	40 x 40 x 110	21	Rounded corner, and the plug that forms the draw-rod nut pocket; bolts to the floor and both walls.

File	What	Qty	Size (mm)	g (all)	Notes
`core-center-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-outboard-L-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-trough-band-L-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-draw-rod-L-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-foot-a-L-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-outboard-R-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-trough-band-R-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-draw-rod-R-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`core-foot-a-R-post.stl`	Pocket core (shell, stays in)	1	12 x 12 x 81	5	Post for a foam core: push it 30 mm into a snug hole in the foam and glue it.
`cast-in-base-front-tube-plug-l.stl`	Cast-in part (stays in)	2	43 x 10 x 43	15	Cast in, stays.
`cast-in-base-front-spigot-sleeve-l.stl`	Cast-in part (stays in)	2	43 x 69 x 43	22	Cast in, stays.
`cast-in-leg-front-l-plug.stl`	Cast-in part (stays in)	2	31 x 31 x 10	8	Cast in, stays.

Uprights (two molds: print quantities are for both)

Folder `print/upright/`, about 313 g.

File	What	Qty	Size (mm)	g (all)	Notes
`kit-back-bracket.stl`	Mold kit (reused)	4	33 x 33 x 23	21	
`kit-front-bracket.stl`	Mold kit (reused)	4	50 x 50 x 22	32	At the top of the front board; its foot sits in the toe filler's lip.

File	What	Qty	Size (mm)	g (all)	Notes
`kit-saddle-flat.stl`	Mold kit (reused)	4	31 x 25 x 25	18	The bridge rests on them; its rods locate it.
`filler-toe-filler.stl`	Filler (removable)	2	80 x 60 x 48	54	Rounds the nose into the slope; the front board's foot sits in its lip. Bolts to the floor.
`former-foot-gasket-groove-part1of4.stl`	Former (removable)	2	68 x 139 x 8	7	Pins into the floor.
`former-foot-gasket-groove-part2of4.stl`	Former (removable)	2	4 x 139 x 8	3	
`former-foot-gasket-groove-part3of4.stl`	Former (removable)	2	4 x 139 x 8	3	
`former-foot-gasket-groove-part4of4.stl`	Former (removable)	2	68 x 139 x 8	7	
`plug-slope-plug-lower.stl`	Plug (removable)	2	32 x 61 x 38	19	Forms a nut pocket in the slope; bolts to the front board; its rod stops inside it.
`plug-slope-plug-upper.stl`	Plug (removable)	2	32 x 61 x 38	19	Forms a nut pocket in the slope; bolts to the front board; its rod stops inside it.
`plug-tie-rod-pocket.stl`	Plug (removable)	4	32 x 32 x 49	33	Forms a top nut pocket: slides down its rod; the bridge holds it down.
`kit-plate-spacer.stl`	Mold kit (reused)	4	16 x 16 x 31	15	Between the bridge and the top plate, on its greased 3/8 bolt.
`kit-post-bushing.stl`	Mold kit (reused)	4	80 x 80 x 22	67	Push fit in a 2-1/2 in hole-saw hole in the floor, flange underneath.
`cast-in-post-l1-plug.stl`	Cast-in part (stays in)	4	31 x 31 x 10	16	Cast in, stays.

Beam

Folder `print/beam/`, about 154 g.

File	What	Qty	Size (mm)	g (all)	Notes
`kit-corner-bracket.stl`	Mold kit (reused)	4	50 x 50 x 23	33	
`filler-back-edge-part1of4.stl`	Filler (removable)	1	136 x 40 x 40	19	Rounded back edge; bolts to the top wall.
`filler-back-edge-part2of4.stl`	Filler (removable)	1	136 x 40 x 40	19	
`filler-back-edge-part3of4.stl`	Filler (removable)	1	136 x 40 x 40	19	

File	What	Qty	Size (mm)	g (all)	Notes
`filler-back-edge-part4of4.stl`	Filler (removable)	1	132 x 40 x 40	19	
`kit-tube-locator.stl`	Mold kit (reused)	4	43 x 43 x 51	43	Square end into the tube, stem to the end wall, M4 x 70 through the wall into a nut slid in from inside the tube. Leaves a 16 mm hole in each beam end.
`kit-dowel-stop.stl`	Mold kit (reused)	2	18 x 18 x 11	2	Pushes into the dowel hole from outside so the dowel stands 10 mm proud.

Machine parts

Folder `print/machine/`, about 99 g.

File	What	Qty	Size (mm)	g (all)	Notes
`part-pocket-cap-l3.stl`	Machine part	4	32 x 52 x 29	32	
`part-chain-mount-fixed.stl`	Machine part	1	25 x 38 x 21	8	
`part-chain-mount-moving.stl`	Machine part	1	37 x 123 x 45	18	
`part-sump-filter-frame-part1of2.stl`	Machine part	1	148 x 6 x 92	20	
`part-sump-filter-frame-part2of2.stl`	Machine part	1	148 x 6 x 92	20	